

Financial Risk

Market Risk

change in value of portfolio due to changes in mv of assets held

STEPS:

- identify + **define** market risk
 - ↳ variance over specified period
 - ↳ max loss with certain probability $\rightarrow$ VaR
- develop techniques / **models** to quantify risk
 - ↳ calculate risk at point in time
 - ↳ Assumption: Returns $\sim N$
 - ↳ not always suitable
 - 1] fat tailed
 - 2] overly skewed

modeling to measure market risk

1. under mean-var portfolio theory

- non-N in dist
- given $E(\text{return})$, var, cov + portfolio weights for each asset class
 - portfolio ave return + $\sigma \rightarrow \therefore$ can calculate risk measure

2. stochastic model

- project portfolio returns + levels

3. VaR to measure risk of future outcomes

- E(loss) with certain confidence level over certain period of time

4. sensitivity analysis Δ input parameters

assess impact of Δ on risk exposure

Credit Risk

counterparty to an agreement is unwilling or unable to fulfil obligations

- creditworthiness $\leftarrow$ limit credit rating
- clearing house on exchange
- aggregate exposure
- exposure to counterparty $\leftarrow$ hedge: credit derivatives

Operational Risk

loss due to internal fraud or mismanagement

⊗ difficult to quantify + measure

Failed systems, processes or people

- dominance risk
- staff training + understanding

CHAPTER 13: RISK CONTROL

ability to raise funds at appropriate prices

Liquidity Risk

not having sufficient cash to meet operational needs at all times

CF from assets to meet liabilities

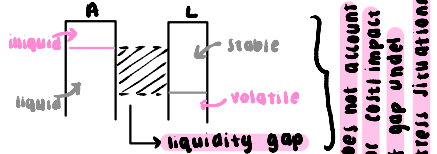
↳ not $\Delta L \rightarrow$ solvency

Gap Analysis

Balance sheet model $\leftarrow$ liquidity policies operating limits

Assets $\leftarrow$ liquid $\rightarrow$ mature < 6 months

Liabilities $\leftarrow$ stable $\rightarrow$ due < 6 months



⊗ Allow for liquidation costs

- brokerage + investment banking fees
- bid-offer spread
- time available for conversions

Relative Performance Risk

underperforming comparable institutional investors

⊗ same technique as market risk

OUT: relative performance

- not absolute / whole market relativity

• minimise $\leftarrow$ commercial matching index tracking

• difficulty $\leftarrow$ appropriate peer group to compare
reliable + accurate data on competitors
allowance for risks of position taken

Duration Analysis

consider impact of change in market conditions

STEP 1:

- calculate PV of A + L under cost of funds rate as discount rate

• Basic gap = $PV_A - PV_L$

- measure $\uparrow$ int rate at all durations
- ↳ $LRE = \text{rate of } \Delta$

STEP 2:

- measure Δ in mv of LRE from Δ in cost of funds

$LRE = 0 \rightarrow$ no liquidity risk

$LRE < 0 \rightarrow$ duration $A > L$

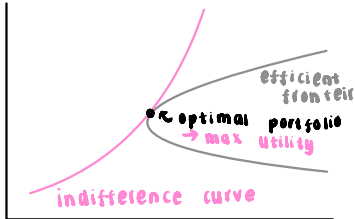
$\uparrow$ int rate = liquidity problem under stress situations

pay to $\leftarrow$ shorten A / lengthen L

Portfolio Theory

Specifies a method to construct a portfolio giving max E(returns) for specified level of risk or min risk for specified return

LIMITATION: ignores investor's liabilities



MVPT application to portfolio selection

- definition of properties of portfolios available
 - ↳ opportunity set
- determine how to choose one out of all feasible portfolios

assumptions of MVPT:

- All E(returns), var + cov of pairs of assets are known
- Investors make decisions on basis of E(returns) + var
- Investors are non-satiated
- Investors are risk averse
- Fixed single step time period
- No taxes / transaction costs
- assets can be held in any amount - no max investment limit
 - ↳ short-selling is permitted
 - ↳ infinitely divisible holdings

Difficulties:

- complexity + data requirements of utility
- utility = subjective
- models not widely used

Factors influencing investment decisions:

- E(returns)
- variance
- suitability of A for L
- marketability
- higher moments of distribution of returns
 - ↳ skewness
 - ↳ kurtosis
- taxes + investment expenses
- restrictions imposed by legislation
- restrictions imposed by trustees
- length of time horizon

Asset-Liability Modelling

objectives stated with ref to Assets Liabilities

PROCESS:

- Construct a model
 - ↳ project A proceeds + L outgo
- examine outcome of investment strategy with model
 - ↳ compare to objectives
- adjust strategy
- repeat until optimal strategy is reached

② Stochastic / deterministic model

↳ consistent assumptions

Advantages: SRM

- encourage formulation of explicit objectives
 - ↳ quantifiable + measurable performance targets
 - ↳ defined horizons
 - ↳ quantified confidence levels for achieving target
- helps refine objectives → feedback
- success monitored by regular valuations

Risk-Free Rate

rate at which money is borrowed + lent and there is no credit risk

→ money certain to be repaid

- ① Treasury Bill
 - ② Government Bond
- ↑ national debt

only valid in closed-economy

- all financial transactions in currency of that economy

Print money to pay debt not valid with:

- sharing currency
- tied exchange rate closely to something else

key measure:

- Ratio of outstanding debt to GDP

- maturity of existing debt

1. cost of servicing debt
2. raise financing cap as debt matures